



PROFILE

PhD candidate in Computer Science at ETH Zurich working on LLM agents, world models, foundation models, and multi-modal AI. Experienced in web-scale data curation, self-supervised pretraining, multi-modal reasoning, and LLM post-training, with end-to-end experience in data pipelines, distributed training, and benchmark development. Research spans computer vision, natural language processing, and their applications in healthcare, with publications in leading venues including Nature Machine Intelligence, ACL, EMNLP, ICLR, MICCAI and ECCV.

EDUCATION

ETH Zurich, Zurich, Switzerland <i>Ph.D., Dept. of Computer Science; ETH AI Center Doctoral Fellow (<1% acceptance rate)</i>	06.2023 – present
ETH Zurich, Zurich, Switzerland <i>M.Sc., Dept. of Information Technology and Electrical Engineering</i>	09.2020 – 03.2023
Xi'an Jiaotong University, Xi'an, China <i>B.Eng. in Electrical Engineering</i>	09.2016 – 06.2020

RESEARCH EXPERIENCE

ETH AI Center, Zurich, Switzerland <i>Doctoral Fellow</i>	06.2023 – Present
• Swiss AI Initiative on Egocentric World Models (ECCV 2026) ◦ Scalable Egocentric Data Engine: Built an automated 3D hand annotation and spatiotemporal refinement pipeline over Ego4D and EgoDex, producing million-scale high-quality egocentric video clips with temporally consistent 3D hand trajectories for large-scale world-model training; ◦ 3D-Controllable World Model: Developed a controllable video generation framework with occlusion-aware feature propagation and 3D geometric embeddings for precise, structurally consistent hand-object interaction generation; ◦ Parameter-Efficient Fine-tuning: Adapted WAN 2.1 with LoRA and a lightweight 3D control module, enabling fine-grained interactive manipulation from individual finger joints to full hand trajectories; ◦ Cross-Embodiment Generalization: Achieved state-of-the-art video fidelity and 3D motion control on Ego4D and EgoDex, reducing FVD by $\geq 16\%$ and 3D hand MPJPE by up to 68%, with cross-embodiment transfer from human interactions to dexterous robotic hands. • Swiss AI Initiative on Healthcare Foundation Models (MICCAI 2025) ◦ Web-Scale Data Curation: Built a dataset of ~350 million images through web-scale data collection and curation; designed hierarchical-clustering-based data curation and cluster-stratified sampling for data-efficient pretraining; ◦ Self-Supervised Pretraining: Scaled pretraining to 128 NVIDIA GH200 GPUs using DINOv2 framework; built reusable distributed training and high-throughput data-loading infrastructures for foundation model training; ◦ Foundation Model Benchmarking: Built a multi-scale downstream evaluation pipeline; demonstrated curated pretraining on 10% data outperformed full-data pretraining on various downstream tasks.	
MBZUAI, Abu Dhabi, UAE <i>Visiting Researcher</i>	10.2025 – 01.2026
• Multi-modal LLM for Complex Clinical Reasoning (ACL 2026) ◦ Multi-modal Data Curation: Designed an LLM-assisted data pipeline with human-in-the-loop filtering, quality control, and expert validation for clinical cases collected from online medical journals; ◦ Scalable Model Serving & Evaluation: Built a unified evaluation framework integrating GPT/Claude/Gemini APIs and vLLM-served open-source LLMs (2B–78B); benchmarked 18 models on complex clinical reasoning tasks; ◦ Failure Mode Analysis: Developed systematic model diagnostics using <i>Relative Attention per Token</i> and <i>Evidence Sensitivity</i>; identified the cross-modal clinical evidence utilization gap rooted in inherent modality biases; ◦ SFT Data Recipe Design: Designed a Evidence-Sensitivity-guided SFT data recipe that up-samples systematically underutilized modalities; achieved 2× the downstream task accuracy gain compared to general SFT using Qwen2.5-VL (7B). • LLM Agents for Sequential Clinical Decision Making (EMNLP 2026)	

- **Clinical Agent Sandbox Design:** Engineered an interactive clinical sandbox with persona-driven patient simulation, stateful multi-turn interactions, and dynamically revealed observations, modeling partial observability and heterogeneous patient behavior across 2,428 expert-validated medical records;
- **Patient-Doctor Agent Evaluation:** Built a multi-turn rollout evaluation framework for interactive agents, with trajectory-level reward collection, task-specific grading rubrics, and automated LLM-as-a-Judge scoring;
- **Real-World Complexity Action Space with Safety Abstention:** Formulated a Constrained Markov Decision Process (CMDP)-based agent environment with structured action outputs, state-dependent action masking, iterative information gathering, and safety-aware abstention to model realistic sequential decision-making;
- **Trajectory-Level Agent RL:** Fine-tuned Qwen2.5 (7B) with GRPO using trajectory-level rewards and Lagrangian safety/cost constraints; improved downstream task accuracy by **30%** while reduce safety violations by **8%**.

PROFESSIONAL EXPERIENCE

Apertus, Zurich, Switzerland <i>Machine Learning Engineer</i>	09.2026 – present
• Work on Apertus (v2.0) post-training and evaluation.	
Apple, Shanghai, China <i>Machine Learning & AI Engineer Intern, Greater China R&D</i>	01.2026 – 06.2026
• Work on LLM agents for failure analysis.	
IBM Research, Rueschlikon, Switzerland <i>Machine Learning Research Intern, Accelerated Discovery & AI</i>	09.2021 – 04.2022
• Work on semi-supervised learning for large-scale image classification. (ECCV 2022)	

SELECTED PUBLICATIONS

[1] GPAGentBench-2K: Benchmarking Large Language Model Agents in Complex Clinical Action Space <i>B. Chen, X. Liu, Y. Ao, H. Do, J. Qiu</i>	EMNLP 2026
[2] Revisiting Greedy Decoding for Visual Question Answering: A Calibration Perspective <i>B. Chen, X. Liu, Y. Ao, J. Qiu</i>	EMNLP 2026
[3] Controllable Egocentric Video Generation via Occlusion-Aware Sparse 3D Hand Joints <i>C. Zhang*, B. Ye*, B. Chen*, A. Delitzas, F. Wang, M. Pollefeys, X. Wang</i>	ECCV 2026
[4] MEDSYN: Benchmarking Multi-Evidence Synthesis in Complex Clinical Cases for multi-modal Large Language Models <i>B. Chen, X. Liu, J. Peng, M. Frey-Marti, K. Lam, B. Zheng, L. Li, J. Qiu</i>	ACL 2026
[5] YoNoSplat: You Only Need One Model for Feedforward 3D Gaussian Splatting <i>B. Ye, B. Chen, H. Xu, D. Barath, M. Pollefeys</i>	ICLR 2026
[6] Histopathology-based Protein Multiplex Generation Using Deep Learning <i>S. Andani*, B. Chen*, J. Ficek-Pascual, S. Heinke, R. Casanova, et al.</i>	Nature Machine Intelligence
[7] Revisiting Automatic Data Curation for Vision Foundation Models in Digital Pathology <i>B. Chen, C. Vincent-Cuaz, L. A. Schoenpflug, M. Madeira, et al.</i>	MICCAI 2025
[8] Generalizable Single-Source Cross-Modality Medical Image Segmentation via Invariant Causal Mechanisms <i>B. Chen, Y. Zhu, Y. Ao, S. Caprara, R. Sutter, G. Rätsch, E. Konukoglu, A. Susmelj</i>	WACV 2025
[9] Differentiable Zooming for Multiple Instance Learning on Whole-Slide Images <i>K. Thandiackal*, B. Chen*, P. Pati, G. Jaume, D. F. K. Williamson, M. Gabrani, O. Goksel</i>	ECCV 2022
* Equal contribution.	

TECHNICAL SKILLS

Infrastructure & Languages: AWS, Slurm, Docker, Git, Python, C++
Training & Serving: PyTorch, PyTorch Lightning, Transformers, FSDP, vLLM, LangChain, LangGraph
Post-Training & Alignment: SFT, RLHF (DPO, GRPO), OPD, Reward Modeling, Preference Data, Agent RL, Tool-Calling Evaluation
Models: LLMs, Vision-Language Models, Diffusion Models, LLM Agents, CV, NLP

REVIEWER SERVICE

ACL, EMNLP, NeurIPS, ECCV, IJCV, IEEE-TMI, etc.	2023 – Present
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